SUPPLEMENTARY NOTE TO "EXHAUSTIVE SEARCHES FOR WILLIAMS NUMBERS": TWO-SIDED SUBSET PRODUCTS AND AN ONSET HEURISTIC

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Abstract. This note records two heuristic experiments that accompanied the computations of the main paper: a measurement of how the Erdős–Alford–Granville–Pomerance subset-product construction degrades when both $p-1$ and $p+1$ must be controlled, and an Erdős–Pomerance-style estimate of the size at which the first Williams number might appear. Neither is a theorem; both are reported with their assumptions so that they can be compared with Zhang's conjectures and improved upon. Code: `agrawal/` in the repository https://github.com/razv2025/carmichael-lucas-carmichael.

A *Williams number* is a squarefree composite n with $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for every prime $p \mid n$; see the main paper for background, notation and references (cited here by the same keys).

1. Two-sided subset products: a measurement

In the subset-product method one fixes a smooth modulus Λ, harvests primes p with $(p-1) \mid \Lambda$, and looks for a subset with product $\equiv 1 \pmod{\Lambda}$; the expected number of solutions is about $2^{(\text{pool size})-\log_2 \varphi(\Lambda)}$, and the method succeeds because the pool grows faster than $\log_2 \varphi(\Lambda)$. For Williams numbers one needs simultaneously $(p+1) \mid \Lambda'$ and product $\equiv -1 \pmod{\Lambda'}$, and the two moduli interact: an odd prime ℓ cannot divide both some $p_i - 1$ and some $p_j + 1$ (else $n \equiv 1 \equiv -1 \pmod{\ell}$), so every small prime must be assigned to one side, and the 2-adic branch is fixed. Define the *margin* of a configuration as (number of usable primes) $-\log_2 \varphi(\text{lcm of all required } p \mp 1)$, so that the expected number of two-sided solutions is about 2^{margin}.

We measured the margin over three families of configurations: random smooth moduli (3,250 configurations, all with margin > -14 searched exhaustively by meet-in-the-middle, no solution); pools of all primes $p \le X$ with both $p-1$ and $p+1$ y-smooth, with the side assignment optimised by hill-climbing (hundreds of restarts); and pools restricted to primes with few prime factors in $p \mp 1$. The best margins were -8.2 (tiny three-prime systems), $-29.0/-30.6/-31.0/-36.9$ for $y = 60/100/140/200$, and -36.9 for the sparse pools. The wall is flat or worsening in every direction we varied: each usable prime contributes one bit of subset freedom but about 2.5 bits of modulus, because a prime that is useful on one side is almost never useful on the other. This is a measurement, not a theorem. The side-assignment of the small primes was formalised by Zhang [2], who splits the prime factors of the modulus into two groups serving $p-1$ and $p+1$ respectively, conjectures that the number of Williams numbers built from a modulus with s prime factors grows only like $2^{2^{s^{1/2-o(1)}}}$ (against $2^{2^{s(1+o(1))}}$ for Carmichael numbers), and exhibits a set of 1,029 primes expected to contain at least 2^{24} Williams numbers, none of which has been found. Our margins are a small-scale measurement consistent with Zhang's conjecture and with the "intractable" judgement of [1].

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2. A heuristic onset estimate

This appendix is heuristic. For comparison, Zhang [2, Remark 7.4] predicts that a Williams number built from his set of 1,029 primes would have about 428 prime factors; the estimate below concerns the smallest Williams numbers of any origin and gives fewer, but likewise many, prime factors. An Erdős–Pomerance-style count of k-factor products of primes with $p \mp 1$ both y-smooth, including the side-consistency factor $2(1-1/\ell)^k - (1-2/\ell)^k$ for each odd prime $\ell \le y$ and a factor 2^{1-k} for the 2-adic branch, calibrated on the one-sided side against Pinch's counts $C(10^{15}) = 105{,}212$, $C(10^{18}) = 1{,}401{,}644$ and on the two-sided side against our measured counts of twin-smooth primes, places the first Williams number near 10^{208}–10^{490} with $k \approx 80$–110 and $y \approx 40$–90, and the first Lenstra–Pomerance counterexample to Agrawal's conjecture near 10^{2400}–10^{2900} with $k \approx 270$. The ranges reflect the sensitivity to the per-prime calibration; the qualitative conclusion—far beyond exhaustive search, reachable only by a construction that circumvents §**??**—is robust. The same machinery places the first Carmichael number near 10^3 (truth: 561).

References

[1] R. Baillie, A. Fiori, S. S. Wagstaff Jr., Strengthening the Baillie–PSW primality test, *Math. Comp.* 90 (2021), 1931–1955.
[2] Z. Zhang, Estimating the counts of Carmichael and Williams numbers with small multiple seeds, *Math. Comp.* 84 (2015), 309–337.
[3] R. G. E. Pinch, Absolute quadratic pseudoprimes, Turku 2007.

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